

Models, applications, and limitations of the responsible adoption of big data and artificial intelligence in public policy



José David Vallejo Manzur^a  | Erick Leobardo Alvarez-Aros^b  

^aThe College of Tamaulipas, Victoria City, Tamaulipas, Mexico.

^bGraduate School, Catholic University of El Salvador, Santa Ana, El Salvador.

Abstract Big data and artificial intelligence are transforming the way governments design, implement, and evaluate public policies. These technologies offer unprecedented capabilities for analyzing complex datasets, anticipating social trends, and allocating resources more effectively. However, their integration into the public sector presents critical challenges related to institutional inertia, data ethics, technical capacity, and governance structures. This article presents a systematic review of the adoption of Big Data and AI in public policy, examining theoretical models, practical applications, and structural barriers. The study employs a PRISMA protocol, synthesizing 103 peer-reviewed articles and institutional reports published between 2010 and 2024. It draws on established frameworks such as Rogers' Diffusion of Innovations, the Technology Maturity Model, and UTAUT while incorporating emerging concepts from algorithmic governance, sociotechnical systems theory, and digital ethics. The results reveal wide variance in adoption across regions and policy domains. High-income countries such as Estonia and Singapore exhibit mature integration in areas such as urban mobility and e-health, while developing contexts face persistent gaps in infrastructure and talent retention. Additionally, the analysis highlights the interplay between predictive modeling and policy foresight, emphasizing how AI and big data are reshaping the temporality of governance—from reactive management to anticipatory decision-making. The analysis also highlights the role of human–AI interaction, including behavioral dynamics such as automation bias, and emphasizes the importance of legitimacy and citizen trust in shaping adoption trajectories. The review identifies three areas for future research: (1) the development of interoperable and ethical-by-design platforms; (2) the measurement of social equity impacts in data-driven governance; and (3) the adaptation of global technological standards to local institutional capacities. It concludes an integrated framework for responsible AI and Big Data adoption in public administration, grounded in transparency, equity, and relevance. It combines technical and institutional perspectives while emphasizing accountability, fairness, and citizen acceptance.

Keywords: e-governance, emerging technology, industry 4.0, industry 5.0, digital government, algorithmic governance

1. Introduction

In the current era of accelerated digital transformation, governments worldwide face mounting pressure to address increasingly complex, dynamic, and interdependent societal challenges. From climate change and global pandemics to demographic transitions and economic instability, public administrations are required to make high-stakes decisions under conditions of uncertainty, scarcity, and rapidly evolving citizen expectations. In this context, the integration of big data and artificial intelligence (AI) has emerged as a pivotal enabler of strategic governance, offering new capabilities for real-time monitoring, predictive modeling, and anticipatory decision-making (OECD, 2019; Kitchin, 2014).

Big data enables the capture and analysis of vast, heterogeneous, and high-velocity datasets from multiple sources—ranging from administrative records and digital transactions to IoT devices and social media streams (Laney, 2001; Mayer-Schönberger & Cukier, 2013). When coupled with AI techniques such as machine learning, deep neural networks, and natural language processing, these data streams can be transformed into actionable insights that support public service delivery, policy formulation, and performance evaluation (Goodfellow et al., 2016; Devlin et al., 2019). The shift from intuition-driven to data-driven policy design represents a profound change in the epistemological foundations of public administration (Margetts & Dunleavy, 2013).

However, the integration of Big Data and AI into public policy is not merely a technical undertaking. It involves substantial institutional, cultural, ethical, and regulatory dimensions. While some governments—such as those of Estonia, Singapore, and Denmark—have implemented ambitious national strategies to embed these technologies into core governance functions (Toots et al., 2017; Batty, 2013), others remain constrained by fragmented digital infrastructure, limited technical expertise, and outdated administrative procedures (Baller et al., 2016). Moreover, concerns about algorithmic bias, opacity, surveillance,



and data sovereignty pose significant normative and political challenges to their widespread adoption (Mittelstadt et al., 2016; Crawford et al., 2021).

The academic literature on the intersection of AI, big data, and public policy has grown exponentially over the last decade. Nonetheless, existing studies are often dispersed across disciplinary silos, lack conceptual integration, or focus narrowly on specific applications (e.g., smart cities, e-health, and election analytics). Few efforts have attempted to systematize the theoretical foundations, evaluate comparative adoption patterns, and identify the cross-cutting barriers that condition the ethical and effective use of these technologies in the public sector (Alvarez-Aros & Bernal-Torres, 2021).

This article addresses that gap by conducting a systematic literature review of peer-reviewed research and institutional reports published between 2010 and 2024. This study seeks to answer three guiding questions: (1) Which theoretical models explain the adoption of Big Data and AI in public policy? (2) What practical applications have been documented across different policy domains and regions? (3) What barriers and limitations hinder the responsible integration of these technologies in public governance? By engaging critically with these questions, the article makes two main contributions. First, it proposes an integrative analytical framework that synthesizes existing models of technological adoption, extending them with insights from sociotechnical systems theory, digital ethics, and algorithmic governance and incorporating recent evidence on human–AI behavioral dynamics and citizen acceptance, highlighting how legitimacy, accountability, and trust shape adoption outcomes. Second, it provides a comparative overview of real-world implementations of Big Data and AI in diverse policy areas—including urban planning, public health, environmental sustainability, and social protection—drawing attention to the institutional and contextual factors that shape their outcomes and demonstrates the need to embed accountability mechanisms and fairness considerations into public sector applications.

The article is structured as follows. Section two develops the theoretical framework, defines key concepts and reviews established and emerging models of adoption and governance. Section three outlines the methodology, including the review protocol, inclusion criteria, and analytical procedures. Section four presents the findings, which are organized into three subsections: theoretical models, practical applications, and barriers. Section five discusses the implications, proposes an original conceptual synthesis and identifies future research directions. Finally, Section six offers conclusions and policy recommendations for enhancing the ethical, inclusive, and context-sensitive integration of Big Data and AI in public administration.

2. Theoretical Framework

The integration of big data and artificial intelligence (AI) into public policy is not solely a matter of technological advancement but also a complex transformation rooted in theoretical, institutional, and ethical considerations. To understand how these technologies are adopted, operationalized, and contested within the public sector, it is necessary to examine a set of conceptual frameworks that capture both the drivers and the frictions inherent to digital innovation in governance.

2.1 Definitions and conceptual scope

Big data in the context of public policy refer to large-scale, high-frequency, and high-variety data streams collected from diverse sources—including administrative records, online platforms, sensor networks, and citizen interactions—which are processed via advanced analytics to inform decision-making (Kitchin, 2014; Mayer-Schönberger & Cukier, 2013). The classic “3 Vs” model—volume, velocity, and variety—has since been expanded to include veracity and value, reflecting concerns about data quality and utility (Laney, 2001; White, 2015).

Artificial intelligence encompasses a spectrum of computational techniques, including machine learning, natural language processing, and neural networks, that enable systems to perform tasks traditionally associated with human intelligence—such as pattern recognition, prediction, and decision-making (Goodfellow et al., 2016; Russell & Norvig, 2021). Within the public domain, AI has been applied to automate services, detect fraud, optimize resources, and simulate complex scenarios in health, education, and urban planning (Das & Zhang, 2019).

Together, Big Data and AI form the basis of what has been termed algorithmic governance—a paradigm in which administrative decisions are increasingly shaped by data-intensive, automated systems, often with limited human oversight (Yeung, 2018; Zuboff, 2019). This shift has profound implications for transparency, accountability, and democratic legitimacy.

2.2 Models of technological adoption in the public sector

The literature on technology adoption offers several explanatory models that have been applied, adapted, and critiqued in the context of public administration.

2.3. Rogers’ diffusion of innovations (DOI) model

Rogers (2003) theorizes that innovation adoption follows a predictable process influenced by five attributes: relative advantage, compatibility, complexity, trialability, and observability. This model is particularly useful in public sector analysis

because it emphasizes how organizational culture, peer influence, and communication channels affect the uptake of innovations.

In the case of AI and big data, relative advantage refers to the perceived benefits over traditional decision-making methods—such as speed, efficiency, or accuracy. However, complexity and lack of trialability often deter implementation in bureaucratic settings resistant to experimentation (Venkatesh et al., 2003).

2.4. *Technology acceptance model (TAM) and UTAUT*

The technology acceptance model (TAM) developed by Davis (1989) and its extension into the unified theory of acceptance and use of technology (UTAUT) by Venkatesh et al. (2003) focus on individual users within organizations. They posit that perceived usefulness and ease of use are critical factors in the adoption of new technologies and are moderated by variables such as age, experience, and voluntariness.

Applied to the public sector, these models highlight the importance of training, institutional incentives, and technical support to ensure that public employees not only accept but also competently deploy Big Data and AI systems (Venkatesh et al., 2003).

2.5. *Gartner's technology maturity model*

Gartner's technology maturity model, which is commonly illustrated through the Hype cycle (Gartner, 2019), portrays the trajectory of emerging technologies as they move from initial excitement to eventual stabilization. It identifies five sequential stages: an innovation trigger that generates early attention, a peak of inflated expectations often followed by disappointment, a trough of disillusionment where many projects falter, a slope of enlightenment as practical applications emerge, and finally a plateau of productivity in which technologies achieve widespread and sustained use. In the context of public administration, this framework evaluates institutional readiness and the degree to which AI and Big Data become embedded within governance systems (OECD, 2019).

Governments such as Singapore and Estonia are cited as having reached the optimization stage, where data-driven systems are embedded into strategic decision-making. In contrast, lower-income countries often remain at the pilot or experimentation stages, struggling with scalability and sustainability (Baller et al., 2016).

2.6. *Sociotechnical systems emergence theory*

The sociotechnical perspective posits that technology and society coevolve and that successful digital adoption requires the alignment of technical systems with organizational norms, legal frameworks, and user capabilities (Hughes, 1987; Margetts & Dunleavy, 2013). This view challenges deterministic narratives by foregrounding the role of human agency, institutional design, and feedback loops. In the realm of public policy, sociotechnical analysis helps explain why AI systems that function effectively in the private sector may underperform or face resistance in the public sphere due to conflicting values, fragmented governance, or political constraints (North & Macal, 2007).

2.7. *Emerging algorithmic governance and data colonialism*

Recent scholarship highlights how algorithmic systems reshape the power dynamics between the state and society. Algorithmic governance involves the delegation of discretion to data-intensive, automated processes, often reducing transparency and public scrutiny (Yeung, 2018; Eubanks, 2018). Moreover, data colonialism criticizes the asymmetrical flow of data from citizens to transnational platforms and from the Global South to the Global North, raising concerns about digital sovereignty and ethical dependencies (Couldry & Mejias, 2019; Crawford et al., 2021). These frameworks call for a critical approach to AI and Big Data adoption, one that interrogates not only their utility but also their implications for equity, agency, and democracy.

2.8. *Emerging ethical and regulatory challenges*

The rapid integration of Big Data and AI into public governance has outpaced the development of corresponding regulatory frameworks. Key ethical challenges include (1) algorithmic bias and discrimination, especially when historical data encode social inequalities (Angwin et al., 2016; Mittelstadt et al., 2016); (2) opacity ("black box" algorithms), which impedes transparency and due processes (Burrell, 2016); (3) privacy erosion, especially in the absence of data minimization and consent mechanisms (Zicari, 2014); and (4) lack of accountability, as decisions become diffused across human-machine assemblages (Floridi et al., 2018). In response, scholars have proposed "ethics-by-design" frameworks and "algorithmic impact assessments" as mechanisms to mitigate risks while preserving the benefits of innovation (Morley et al., 2020; OECD, 2021).

3. **Materials and Methods**

To ensure analytical depth and conceptual validity, this study employed a systematic literature review (SLR) guided by the PRISMA protocol (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) adapted for social science and

public policy research (Moher et al., 2009; Petticrew & Roberts, 2006). This method was chosen to synthesize existing knowledge on the adoption of Big Data and artificial intelligence (AI) in public administration, identify key theoretical frameworks, document empirical applications, and uncover structural barriers.

The review aimed to answer three core research questions: (1) Which theoretical models are used to explain the adoption of Big Data and AI in public policy? (2) What are the most documented applications across policy sectors and national contexts? (3) What are the primary limitations and challenges encountered in practice? This design enabled the identification of patterns, divergences, and gaps across empirical and conceptual studies.

The search was conducted between February and April 2025 across the Web of Science and Scopus databases. Keywords used in Boolean combinations included “big data” AND “artificial intelligence” AND “public policy”; “AI adoption” AND “government” AND “technology governance”; “data-driven policy” OR “algorithmic governance”; and “digital government” AND “policy foresight”. No language restrictions were applied at the initial stage, but only English-language articles were retained for analysis. The references were managed via Mendeley.

The selection of sources was guided by the following inclusion criteria: peer-reviewed journal articles or official institutional reports; published between January 2010 and April 2024; direct focus on the application, governance, or adoption of Big Data and/or AI in public policy; and clearly articulated methodology and identifiable conceptual contributions.

The exclusion criteria included the following: articles focused exclusively on private-sector use of AI/big data; non-peer-reviewed opinion pieces, blog posts, or promotional content; and studies without empirical data or theoretical grounding.

A total of 427 records were initially retrieved. After duplicate removal, title and abstract screening, and full-text evaluation, 103 studies were retained for inclusion. The process of the adapted PRISMA framework is summarized in Table 1.

For the data extraction and coding procedure, a matrix was developed in Excel and NVivo to categorize studies by geographic focus (Global North/South, specific country); policy domain (health, mobility, environment, education, etc.); methodology (qualitative, quantitative, mixed-method); theoretical framework (if any); type of technology applied (big data, AI, both); implementation level (pilot, institutionalized, optimized); and identified barriers or enabling factors. This categorization allowed for comparative analysis and thematic synthesis, and a descriptive summary of the reviewed studies is provided in Table 2.

Table 1 PRISMA-based literature selection summary.

Stage	Records Identified	Records Excluded	Records Retained
Initial search (6 databases)	427	—	—
After duplicate removal	389	38	351
After title & abstract screening	351	218	133
After full-text eligibility review	133	30	103

Source: Authors’ own compilation based on the PRISMA protocol.

Table 2 Summary of reviewed studies by region, policy domain, and methodology.

Region	Number of Studies	Dominant Policy Areas	Common Methodologies
North America	21	Public health, smart cities	Quantitative, case study
Europe	28	Urban mobility, e-government	Mixed-method
Asia-Pacific	19	Education, social protection	Qualitative, surveys
Latin America	12	Environmental policy, social programs	Case study
Africa	9	Health systems, infrastructure	Qualitative
Cross-regional/global	14	Ethical frameworks, adoption models	Literature review

Source: Authors’ synthesis based on the reviewed sample (n=103).

For the analytical approach, the selected literature was analyzed via thematic coding, allowing for inductive category construction related to theoretical underpinnings, technological maturity and readiness, human and institutional capacity, ethical and governance challenges, and reported outcomes and lessons learned. Saturation was reached once no new codes emerged from the last 15 reviewed articles, confirming the sufficiency of the sample.

4. Results

This section presents the main findings of the systematic literature review, which is structured around three key areas: theoretical models of technology adoption (Section 4.1), practical applications across policy domains (Section 4.2), and the structural, organizational, and ethical barriers that hinder effective implementation (Section 4.3). These results reflect a comprehensive synthesis of 103 peer-reviewed publications and institutional reports analyzed via thematic coding and comparative frameworks. The final subsection (Section 4.4) identifies priority research gaps and outlines future lines of investigation aimed at advancing both the academic literature and public governance practice.

Together, these findings offer a multidimensional understanding of how Big Data and artificial intelligence are being adopted, integrated, and contested within public policy settings globally. The diversity of case studies, theoretical interpretations, and operational challenges underscore the need for a context-sensitive and ethically grounded approach to digital transformation in the public sector.

4.1. *Models of technological adoption*

Understanding how governments adopt and integrate emerging technologies such as big data and artificial intelligence (AI) requires more than a descriptive account of tools and outcomes. This necessitates the use of explanatory models that reveal underlying institutional dynamics, organizational incentives, behavioral determinants, and systemic limitations. This section synthesizes the key theoretical models identified in the literature—Rogers’ Diffusion of Innovations (DOI), the Technology Acceptance Model (TAM/UTAUT), and Gartner’s Technology Maturity Model—highlighting their analytical utility, limitations, and empirical applications in the public sector.

4.1.1. *Rogers’ diffusion of innovations (RDOI)*

Rogers’ model remains one of the most cited frameworks for understanding how innovations spread across social systems (Rogers, 2003). The model classifies adopters into five categories—innovators, early adopters, early majority, late majority, and laggards—and identifies five innovation attributes that influence adoption: relative advantage, compatibility, complexity, trialability, and observability. In public sector contexts, relative advantage (e.g., predictive accuracy of AI models) and compatibility (e.g., alignment with existing IT infrastructure) have been shown to accelerate uptake. However, the model is limited in explaining institutional inertia, legal constraints, and ethical concerns that characterize bureaucratic environments (Venkatesh et al., 2003; OECD, 2019). Empirical evidence: Estonia and Singapore exemplify early adoption profiles, leveraging AI for e-governance, mobility, and citizen service platforms (Toots et al., 2017; Batty, 2013). In contrast, many Latin American and African nations fall into the late majority or laggard categories due to infrastructure limitations and policy fragmentation (Baller et al., 2016).

4.1.2. *TAM and UTAUT models*

The Technology Acceptance Model (TAM), introduced by Davis (1989) and its extension into the Unified Theory of Acceptance and Use of Technology (UTAUT) by Venkatesh et al. (2003), examines adoption primarily from the standpoint of individual users within organizations. These frameworks argue that the willingness to adopt a new technology depends on perceptions of usefulness and ease of use but is also shaped by broader contextual elements such as social influence and the presence of organizational support structures (Davis, 1989; Venkatesh et al., 2003; North & Macal, 2007; Morley et al., 2020; OECD, 2019). In public administration, these insights are particularly relevant for understanding how employees respond to AI-based systems, including decision-support tools, fraud detection models, and automated eligibility assessments. Successful implementation in these contexts is strongly associated with adequate training, leadership commitment, and appropriate incentives. Evidence from Japan’s integration of AI in long-term care policy illustrates this point, showing how national training programs enhanced perceptions of usefulness among bureaucrats and frontline staff (OECD, 2019).

4.1.3. *Gartner’s technology maturity model*

Gartner’s maturity framework (commonly represented through the Hype cycle) depicts the evolution of emerging technologies as they move across five phases: an initial innovation trigger, a peak of high expectations, a subsequent period of disillusionment, a stage of gradual enlightenment, and finally a plateau of productivity where adoption becomes stable (Gartner, 2019). This model is widely used by governments and consulting agencies to benchmark institutional progress in digital transformation. In public policy, the model has been applied to assess AI deployment in transportation, public health, and urban analytics. It is particularly useful for identifying institutional bottlenecks, such as insufficient governance structures, fragmented data ecosystems, or overhyped pilot projects that fail to scale. Empirical evidence: Brazil’s National Policy on Digital Government, which includes the use of machine learning to monitor deforestation and infrastructure planning, is currently assessed as being in the “Defined” to “Managed” stages, reflecting intermediate maturity (Amaral et al., 2020). Table 3 presents a comparison of technology adoption models in public policy.

With respect to synthesis and analytical contributions, each of these models offers partial yet complementary insights. While Rogers is effective in capturing institutional diffusion and categorizing adopters, TAM/UTAUT clarify the user-level factors critical in bureaucratic adoption. Gartner’s model adds a systemic lens to assess the longitudinal evolution of technological readiness. However, none of these models alone sufficiently address the ethical, legal, and sociopolitical complexities that emerge when predictive algorithms and massive datasets are embedded in policy decisions. As such, an integrative approach—one that combines behavioral, institutional, and maturity perspectives with critical frameworks from algorithmic governance and digital ethics—is required to assess and guide responsible adoption.

Table 3 Comparative overview of technology adoption models in public policy.

Model	Key Dimensions	Strengths	Limitations	Example Countries
Rogers' Diffusion of Innovations TAM/UTAUT	Relative advantage, compatibility, complexity, trialability, observability Perceived usefulness, ease of use, social influence, facilitating conditions	Explains adoption patterns across institutions; accounts for system-level diffusion Highlights user behavior, training needs, and incentive alignment	Understates political resistance, legal constraints, and ethics Ignores systemic/organizational factors; limited to microlevel	Singapore (early adopters); Ghana (laggards) Japan (AI in aging policy); UK (service automation)
Gartner's Technology Maturity Model	Innovation trigger → Plateau of productivity	Benchmarks institutional progress; adaptable to diverse contexts	Lacks behavioral nuance; may be influenced by commercial bias	Brazil (digital governance); Rwanda (early AI in health)

Source: Authors' synthesis is based on Rogers (2003), Venkatesh et al. (2003), and Gartner (2019), and case studies are reviewed.

4.2. Practical applications of big data and AI in public policy

While theoretical models help explain the dynamics of adoption, practical applications provide critical insights into how big data and artificial intelligence (AI) function in real-world governance contexts. Across sectors and regions, these technologies have been used to address complex public challenges, from disease surveillance and traffic optimization to fraud detection and environmental protection. This section synthesizes prominent use cases documented in the literature, organized by policy domain.

With respect to urban mobility and infrastructure, in densely populated cities, AI and big data have enabled dynamic traffic management, predictive maintenance of infrastructure, and adaptive public transportation systems. For example, Singapore employs real-time mobility analytics and AI-enabled congestion pricing to reduce travel time by 25%, integrating IoT data from sensors and public transit usage patterns (Batty, 2013; OECD, 2019). For example, London has used big data from GPS-equipped buses and traffic sensors to optimize bus routing and reduce carbon emissions by 18% (Toots et al., 2017). Moreover, India launched the "Smart Cities Mission," which uses AI for traffic forecasting and infrastructure planning in cities such as Pune and Bengaluru, leading to a 20% reduction in commuting delays (Kitchin, 2014).

With respect to public health and epidemiology, AI has proven particularly useful in outbreak prediction, health service allocation, and population health management. For example, Rwanda implemented AI-based malaria prediction models with 92% accuracy, enabling early intervention and improved medicine distribution in rural areas (Das & Zhang, 2019). For example, during the COVID-19 pandemic, South Korea and the United States applied big data analytics to contact tracing, epidemiological modeling, and real-time health resource management (Choi & Varian, 2012; OECD, 2021). In other examples, Japan has used Big Data to model long-term care demands due to population aging, influencing pension and eldercare policy adjustments (OECD, 2019).

With respect to environmental sustainability and climate policy, Big Data and AI support environmental forecasting, resource management, and climate adaptation strategies. For example, Brazil employs satellite data and machine learning to monitor Amazonian deforestation in near real time, reducing forest loss by 15% annually (Amaral et al., 2020). For example, Canada uses AI to predict wildfires and coordinate early evacuation plans, mitigating economic and human losses by 22% in affected regions (Manyika et al., 2011). For example, the Netherlands integrates AI with hydraulic models for flood forecasting and water management, reducing infrastructure damage during peak rainfall (Hardoon & Shmueli, 2013).

With respect to social protection and inclusion, data-driven tools enable targeted subsidies, fraud detection, and identification of at-risk populations. For example, in Denmark, AI analyzes household energy use to identify low-income communities at risk of energy poverty, leading to tailored subsidies that improve energy access (Hardoon & Shmueli, 2013). For example, India's Aadhaar system combines biometric Big Data with algorithmic controls to reduce leakage in welfare schemes and improve targeting efficiency (Muralidharan et al., 2016). Example 3: Chile's Sistema de Protección Social integrates social and administrative databases to tailor support programs on the basis of predictive vulnerability scoring (OECD, 2019).

With respect to education and human capital development, AI supports personalized learning and system-level analytics in public education systems. For example, in Singapore, AI-powered platforms support individualized learning trajectories and early intervention for at-risk students, reducing dropout rates by 12% (Kitchin, 2014). For example, USAID piloted AI-based literacy diagnostics in rural Kenya and Bangladesh, improving early learning outcomes through adaptive feedback mechanisms (UNESCO, 2021). For example, in Uruguay, the Ceibal platform integrates big data analytics to track national student progress and tailor teacher training in real time (OECD, 2021). The details are presented in Table 4.

With respect to cross-cutting lessons, despite sectoral differences, the literature reveals seven recurring factors: (1) political commitment and strategic vision (e.g., Singapore, Canada); (2) investments in digital infrastructure and open data platforms; (3) capacity building for public employees and local technical communities; and (4) legal frameworks allowing secure,

ethical data use. Conversely, failures or underperformance are associated with (5) fragmented systems and a lack of interoperability; (6) limited human capital in AI/data science; and (7) inadequate regulation around privacy and bias. These insights support the argument that technology alone does not drive transformation; rather, the institutional context, governance design, and stakeholder engagement are equally decisive.

Table 4 Practical applications of AI and Big Data in public policy by country and sector.

Country	Policy Domain	Technology Used	Application Description	Documented Impact
Singapore	Urban mobility	AI + IoT	Congestion pricing, route optimization	25% reduction in travel time
London (UK)	Public transport	Big Data	Dynamic bus routing via GPS data	18% emission reduction
Rwanda	Health (malaria)	AI	Early outbreak prediction and drug allocation	92% prediction accuracy
Brazil	Environmental monitoring	Big Data + ML	Deforestation detection via satellite imagery	15% annual reduction in forest loss
Canada	Disaster response	AI	Wildfire forecasting and evacuation coordination	22% lower impact in affected zones
Denmark	Energy policy	AI	Targeted subsidies for energy-poor households	Improved subsidy targeting
Japan	Aging and health policy	Big Data + AI	Long-term care demand modeling	Policy adjustments in pensions
India	Welfare programs	Big Data + Biometric AI	Leak detection in Aadhaar-enabled benefit transfers	Reduced fraud, improved targeting
Singapore	Education	AI	Personalized instruction platforms in schools	12% reduction in dropout rates
Chile	Social inclusion	Big Data	Predictive targeting in social safety nets	Increased precision of support

Source: Compiled by the authors from reviewed studies (Das & Zhang, 2019; Amaral et al., 2020; OECD, 2019, 2021; Choi & Varian, 2012; Manyika et al., 2011).

4.3. Barriers and limitations

While the transformative potential of big data and artificial intelligence (AI) in public policy is increasingly acknowledged, their adoption is constrained by a complex set of structural, institutional, technical, and ethical barriers. This section synthesizes the most frequently cited limitations in the reviewed literature, organized into four dimensions: technological infrastructure, human capital, data governance and ethics, and organizational dynamics. These barriers vary across regions and development levels, often reinforcing existing asymmetries in state capacity and digital sovereignty.

Technological infrastructure and interoperability. In many countries, especially in the Global South, the foundational digital infrastructure needed to deploy AI and big data tools is limited or unevenly distributed. Challenges include (1) fragmented databases and legacy IT systems that prevent interoperability across agencies (OECD, 2019; Baller et al., 2016); (2) low bandwidth and connectivity in rural or marginalized areas; (3) the absence of real-time data systems, hindering timely decision-making; and (4) vendor lock-in and dependence on proprietary foreign platforms, limiting adaptability and innovation (Crawford et al., 2021). These limitations not only reduce system efficiency but also restrict long-term scalability.

Human capital and institutional capabilities. Even in technologically advanced countries, public administrations often lack professionals with adequate expertise in data science, machine learning, and algorithmic auditing (Margetts & Dunleavy, 2013; OECD, 2021). Key issues include talent shortages in public sector data roles, brain drain to the private sector due to higher compensation and prestige, limited training programs for public servants on emerging technologies, organizational resistance to digital skill development among older cohorts, and capacity building, which remains one of the most frequently cited prerequisites for successful AI integration in government (Bernal-Torres et al., 2025).

Data governance, privacy, and algorithmic bias. The ethical and legal dimensions of Big Data and AI adoption are increasingly subject to scrutiny. Risks include a lack of data protection legislation or enforcement capacity; opacity in algorithmic decision-making, hindering accountability (Burrell, 2016); embedded biases in training datasets, leading to discriminatory outputs (Angwin et al., 2016; Mittelstadt et al., 2016); and the exclusion of vulnerable populations due to data invisibility or digital illiteracy. These challenges are particularly acute in contexts with weak democratic institutions or unregulated surveillance architectures.

Organizational fragmentation and political barriers. Finally, organizational culture and institutional design significantly shape the success or failure of AI and Big Data initiatives. Common barriers include rigid hierarchies and siloed decision-making; a lack of strategic alignment between digital policy and sectoral ministries; short-term political cycles, which undermine continuity in digital transformation; and limited stakeholder engagement, which reduces legitimacy and the adoption of

algorithmic systems (Morley et al., 2020). Without a coherent governance framework and interagency coordination, even well-funded initiatives may fail to achieve systemic impact. Table 5 shows a summary of the barriers to Big Data and AI adoption by region.

Table 5 Summary of barriers to Big Data and AI adoption by region and category.

Region	Technological Barriers	Human Capital Gaps	Ethical & Governance Issues	Organizational Challenges
North America	Legacy systems in government	Shortage of public-sector data scientists	Concerns over surveillance and private sector dominance	Fragmentation between federal/state levels
Europe	Interoperability limitations	Resistance in older bureaucracies	Strong regulation, but gaps in enforcement	Moderate alignment across ministries
Latin America	Infrastructure gaps, rural–urban divide	Low digital literacy in public sector	Weak data protection frameworks	Political instability, low continuity
Asia-Pacific	Varies widely (e.g., strong in Singapore, weak in rural India)	Uneven technical training	Concerns about state-led surveillance	Centralization without transparency
Africa	Severe connectivity challenges	Acute shortage of data specialists	Low regulatory capacity	High dependency on international donors
Global	Vendor lock-in, poor scalability	Brain drain to private sector	Algorithmic opacity and bias	Institutional silos, poor coordination

Source: Authors' synthesis from OECD (2019, 2021), Mittelstadt et al. (2016), Baller et al. (2016), Crawford et al. (2021), and Amaral et al. (2020) and reviewed case studies.

Toward barrier-responsive strategies. To address these barriers, the literature suggests several enabling strategies: (1) public–private knowledge partnerships for technical capacity building (e.g., Singapore, Japan); (2) opening data standards and interoperability frameworks to ensure system cohesion; (3) algorithmic impact assessments and participatory audits to mitigate bias and increase transparency (Morley et al., 2020); and (4) national digital agendas with long-term political backing to withstand electoral turnover. Nonetheless, a context-sensitive approach is essential. Solutions effective in high-income countries may not transfer directly to low-resource environments without significant adaptation.

4.4. Future research lines

The rapid evolution of big data and artificial intelligence (AI) technologies has outpaced scholarly and policy-oriented research in several critical areas. While the literature has expanded substantially over the last decade, the review reveals notable gaps that merit further investigation to ensure the ethical, inclusive, and effective integration of these technologies in public governance. This section outlines key emergent research lines, emphasizing areas where empirical, theoretical, and normative inquiry is urgently needed.

Regulatory and ethical frameworks. There is a clear need for interdisciplinary research that examines how national and supranational regulatory regimes can balance innovation with rights-based protection. Despite the proliferation of ethical guidelines, few studies have evaluated their actual implementation or impact on administrative practices (Floridi et al., 2018; OECD, 2021). Future research should explore mechanisms for embedding ethical values in algorithm design; comparative assessments of data protection laws and AI governance standards; and the role of judiciaries, ombudspersons, and civil society in algorithmic oversight.

Interoperability and system integration. While interoperability is frequently cited as a technical challenge, few studies offer concrete blueprints for achieving data harmonization across agencies. Future inquiry should prioritize the design of open standards and federated data architectures, public-sector data lakes and their legal-institutional implications, and case studies of whole-government data governance models.

Social impact and equity. Despite growing concern over the distributive consequences of AI, systematic evaluations of how data-driven policies affect different social groups remain scarce. Researchers should examine whether predictive models reproduce or reduce existing inequalities, the impact of AI adoption on access to services in marginalized communities, and longitudinal studies on citizen trust and digital inclusion.

High-uncertainty and crisis contexts. Few studies have investigated the role of AI and Big Data in crisis governance, including pandemic response, disaster risk reduction, and geopolitical instability. Future research should address the development of resilient, adaptive algorithms for dynamic environments; applications of agent-based modeling in contingency planning; and ethical trade-offs in real-time, high-stakes data use.

Localization and contextual adaptation. Most frameworks are derived from high-income contexts. There is an urgent need to localize technologies to institutional realities in the Global South. Suggested research lines include the design of low-resource AI systems compatible with limited infrastructure; community-led data governance models in indigenous and rural

contexts; and the politics of technological dependency and digital sovereignty. Table 6 shows the proposed research agenda with key themes, questions, and methods.

Table 6 Proposed research agenda: Key themes, questions, and methods.

Thematic Area	Key Research Questions	Suggested Methods
Ethics and Regulation	How are ethical AI frameworks operationalized in public administration?	Comparative legal analysis, case studies
Interoperability	What technical and legal conditions enable cross-agency data sharing?	System architecture review, design science
Social Equity and Inclusion	Do AI-based policies exacerbate or mitigate inequality?	Mixed-method evaluations, impact assessments
Crisis and Uncertainty	How can AI be adapted for governance under conditions of volatility?	Agent-based simulation, real-time modeling
Localization and Sovereignty	What models exist for decolonized and locally adapted AI governance?	Fieldwork, participatory design research

Source: Authors’ elaboration on the basis of review findings and gaps identified in Amaral et al. (2020), OECD (2021), Morley et al. (2020), and Crawford et al. (2021).

Toward a research–policy interface. Finally, the literature reveals a need for stronger links between academic inquiry and policymaking. Research agendas should be coproduced with public institutions, civil society, and technical experts to ensure usability and real-world impact. This also includes the creation of policy sandboxes and living labs for experimentation under controlled risk. Bridging the gap between empirical insight and administrative reform is not only a matter of knowledge translation—it is central to the democratic legitimacy of algorithmic governance.

5. Discussion

The systematic review conducted in this study reveals a rapidly expanding yet still fragmented field of research and practice surrounding the adoption of Big Data and artificial intelligence (AI) in public policy. While a wide range of models, use cases, and governance challenges have been documented, few studies provide integrative explanations of how institutional context, technological maturity, behavioral dynamics, and ethical imperatives interact in shaping implementation trajectories. This section offers a critical synthesis of the findings and articulates a conceptual framework that bridges disparate strands in the literature while also acknowledging the review’s limitations and drawing implications for theory and practice. Figure 1 shows conceptual framework for responsible AI and Big Data adoption in public policy.

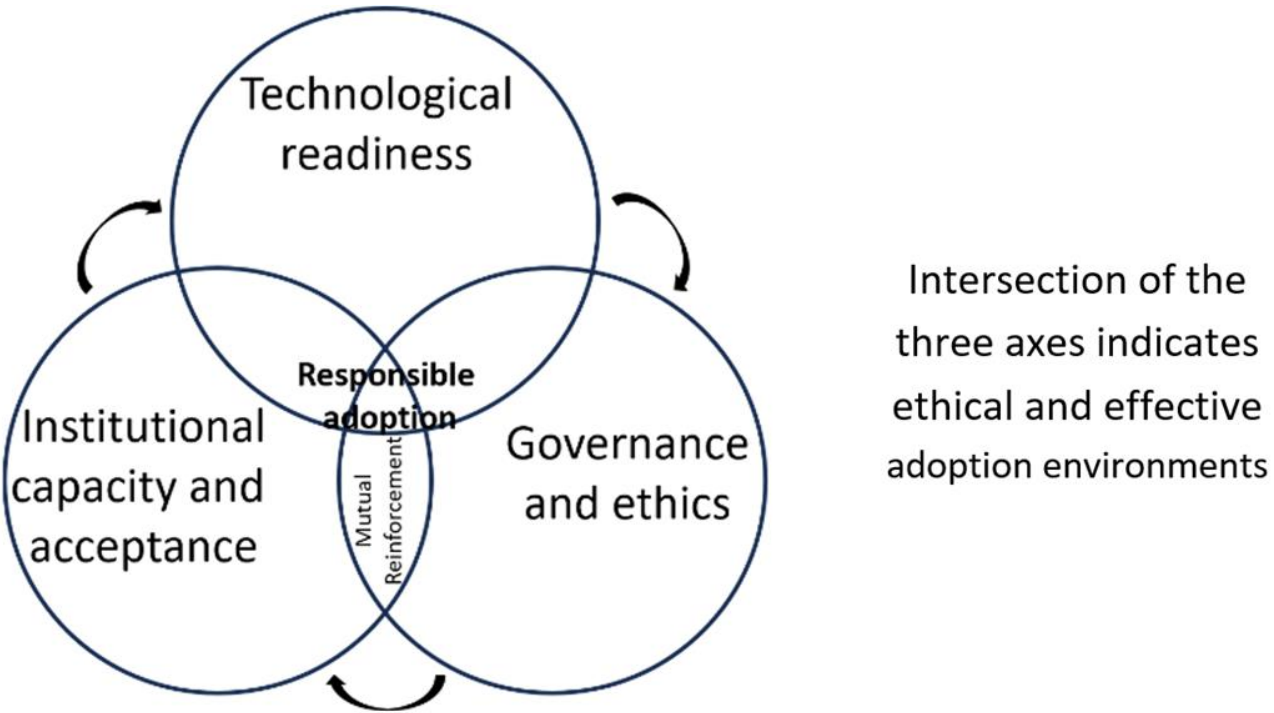


Figure 1 Conceptual framework for responsible AI and Big Data adoption in public policy.

5.1. Patterns and divergences in adoption



Three cross-cutting patterns consolidate across the corpus, each pointing to distinctive mechanisms that shape how Big Data and AI travel from pilots to systemic use in public administration. First, asymmetries in adoption persist across regions and sectors, but these asymmetries are increasingly traceable to governance capacity rather than technical availability alone. Administrations that articulate clear policy problems, establish interagency data arrangements, and build in-house capabilities for model stewardship progress more sustainably than those relying on fragmented procurement or vendor-driven solutions do (Zuiderwijk et al., 2021; Wirtz et al., 2019).

Second, sectoral trajectories diverge according to the maturity of underlying information infrastructures. Domains with long-standing measurement systems—such as urban mobility, epidemiological surveillance, and environmental monitoring—accumulate learning effects more quickly and demonstrate clearer paths from experimentation to routinization (Sun & Medaglia, 2019; Wirtz et al., 2019). In contrast, areas with diffuse objectives or contested values—education, social protection, eligibility determination—face persistent challenges in defining construct validity, mitigating distributional risks, and aligning predictive signals with due processes. Comparative evidence shows that transitions from isolated tools to mission-level reforms depend on managerial sponsorship, embedded change routines, and the translation of performance gains into service-level agreements that matter to line units (Neumann et al., 2022).

Third, a gap remains between instrumental deployments (dashboards, pilots, and discrete decision aids) and strategic transformation (system-wide re-engineering of processes around anticipatory analytics). Bridging this gap requires credible theories of change that link predictive capacity to outcomes that citizens value, together with oversight mechanisms that keep models contestable and auditable at scale (Eyert et al., 2022). Importantly, legitimacy conditions—trust in the deploying authority, perceptions of fairness, and visible redress channels—often determine whether identical tools succeed in one jurisdiction but falter in another (Wenzelburger et al., 2024).

5.2. Tensions between models, practice, and ethics

Classical frameworks—the diffusion of innovations, TAM/UTAUT, and Gartner's maturity curves—explain how technologies spread, how individuals come to use them, and how organizations evolve. However, these models are less equipped to address normative tensions that arise when predictive systems mediate entitlements, sanctions, or rights.

The first tension concerns transparency and accountability. While opening models and datasets has become a governance mantra, empirical work shows that visibility alone does not guarantee contestability if disclosure regimes lack incentives, resources, or enforcement. Registers of algorithms or voluntary impact assessments often remain performative unless coupled with duties to explain, resourced audit capacities, and appeal mechanisms accessible to affected citizens (Ananny & Crawford, 2018; Baykurt, 2022). Regulatory scholars thus emphasize moving beyond transparency toward accountability architectures embedded in routine practices (Eyert et al., 2022).

The second tension arises at the human–AI interface. Behavioral studies demonstrate the risks of automation bias (overreliance on algorithmic outputs) and selective adherence (accepting algorithmic advice when it confirms existing beliefs while rejecting disconfirming signals). These patterns, shaped by workload, interface design, and organizational culture, affect fairness outcomes downstream even when models perform well on technical metrics (Alon-Barkat & Busuioc, 2023). Complementary technical reviews map how bias enters across the lifecycle—data collection, labeling, optimization, and deployment feedback—calling for integrated mitigation strategies that combine technical tools with institutional safeguards (Mehrabi et al., 2021).

Together, these tensions reveal that adoption is not only about diffusion or readiness but also about legitimacy. Responsible use requires frameworks that combine acceptance variables with constructs of trust, fairness, and accountability, ensuring that predictive tools are not only effective but also contestable and democratically legitimate (Neumann et al., 2022; Wenzelburger et al., 2024).

5.3. Toward an integrative framework for public sector AI and big data adoption

The synthesis of theoretical models, empirical applications, and normative debates supports the proposal of an integrative framework along three axes.

- Institutional and technical capability: Robust infrastructures, interoperable data systems, and cross-agency arrangements enable sustainable model lifecycle management. Comparative studies demonstrate that administrations with coherent architectures and shared standards are better able to scale AI deployments beyond pilots (Zuiderwijk et al., 2021; Mergel et al., 2018).

- Human–AI behavioral dynamics: Adoption hinges on how humans interact with algorithms in practice. Designing interfaces to mitigate automation bias, training staff in both technical and ethical competencies, and creating conditions for reflective use are crucial (Alon-Barkat & Busuioc, 2023).

- Governance and accountability: Effective adoption requires transparency mechanisms, algorithmic impact assessments, redress channels, and adaptive regulation. Scholars emphasize that these are meaningful only if tied to enforceable obligations and adequately resourced oversight (Ananny & Crawford, 2018; Eyert et al., 2022).

This tripartite framework extends classical adoption models by embedding behavioral and ethical dimensions. Sustainable adoption emerges not from technical progress alone but from the alignment of capability, behavior, and governance, producing systems that are simultaneously efficient, legitimate, and resilient.

5.4. Limitations of the review

While this review provides a broad synthesis of literature and practice, it is not without limitations. First, it focuses primarily on English-language sources, potentially underrepresenting research and experiences from regions where local scholarship is produced in other languages, such as Latin America or Francophone Africa. Second, the rapid pace of technological change means that the case studies and implementations cited may become outdated within short timeframes, limiting generalizability. Third, the selection of sources emphasized peer-reviewed and institutional publications, leaving out experimental, grassroots, or civil society-driven initiatives that may provide important lessons about AI ethics and citizen engagement.

Moreover, although systematic, the review relies on thematic coding and comparative synthesis rather than on meta-analysis of quantitative findings, which may restrict the statistical generalization of the results. Future research should expand by including multilingual searches, real-time ethnographies, participatory action research, and longitudinal evaluations of adoption trajectories to capture evolving dynamics.

5.5. Implications for theory and practice

The expanded evidence base has significant implications.

For theory, our synthesis highlights the need for hybrid models that link adoption variables (usefulness, diffusion, maturity) with legitimacy constructs (trust, fairness, accountability). Adoption theory thus benefits from integration with critical algorithm studies and behavioral decision research, producing explanations that account for both the mechanics and the politics of algorithmic governance (Baykurt, 2022; Wirtz et al., 2019).

For practice, three priorities are evident:

- Behavior-aware design: Administrations should empirically test for automation bias and selective adherence under realistic workloads and design interfaces that present calibrated uncertainty or counterfactuals to support judgment (Alon-Barkat & Busuioc, 2023).
- Accountability beyond transparency: Registers and guidelines must be tied to enforceable obligations, independent audit capacities, and accessible appeal mechanisms. Symbolic transparency without accountability risks undermining legitimacy (Ananny & Crawford, 2018; Eyert et al., 2022).
- Contextual legitimacy: Trust building and fairness should be treated as design constraints. Clear communication, participatory oversight, and demonstrations of public value are crucial for adoption in high-stakes domains (Araujo et al., 2020; Wenzelburger et al., 2024).

These implications underscore that responsible adoption is a coproduction challenge. Durable value arises when technical capability, human judgment, and institutional accountability reinforce each other. For scholars, this calls for methodological diversification, including experiments, ethnographies, and participatory evaluations. For practitioners, iterative and reflexive governance practices that align innovation with democratic legitimacy are needed.

6. Final Considerations

This section serves as the culmination of the research findings and provides a concise summary of the key outcomes and implications of the study. In this section, the researchers present their final thoughts and insights on the basis of the analysis and interpretation of the data. This approach provides an opportunity to address the research objectives and hypotheses and determine whether they were supported or contradicted by the findings. The conclusions should be supported by evidence from the results and discussion sections, highlighting the significance and novelty of the research outcomes. Additionally, this section may discuss the limitations of the study and suggest potential areas for future research. This section aims to tie together all the threads of the research and provide a clear and coherent summary of the main findings, ultimately contributing to the broader understanding of the research field and potentially influencing future scientific endeavors.

This study systematically examines the state of knowledge and practices surrounding the adoption of big data and artificial intelligence (AI) in public policy. Drawing from over a decade of peer-reviewed research and institutional reports, it has identified the dominant theoretical models, real-world applications, critical barriers, and emerging research agendas in this fast-evolving domain. The findings underscore that while these technologies hold substantial promise for enhancing strategic governance, their adoption is neither automatic nor uniformly beneficial.

6.1. Key conclusions

Adoption is deeply contextual: Technological uptake varies significantly across countries, sectors, and institutional arrangements. High-level infrastructure and national strategies are enabling factors but are not sufficient on their own.

Theoretical models offer partial insights: traditional adoption frameworks (e.g., Rogers, UTAUT, and Gartner) illuminate important dynamics but fall short of addressing ethical, legal, and sociopolitical complexities. An integrative approach is needed.

Real-world applications are growing but uneven: Sectors such as health, mobility, and environmental sustainability lead in experimentation and implementation, whereas others remain underexplored or face systemic barriers.

Barriers are multifaceted: Infrastructural gaps, human capital limitations, data governance weaknesses, and organizational inertia all constrain the effective deployment of AI and big data tools.

The ethical and governance dimensions are underdeveloped: the growing reliance on algorithmic decision-making demands robust regulatory frameworks, transparency mechanisms, and participatory oversight to avoid exacerbating existing inequities.

Future research must be interdisciplinary and action-oriented: To bridge the gap between innovation and public value, researchers must cocreate knowledge with practitioners, policymakers, and affected communities.

6.2. Principles for responsible adoption

On the basis of the evidence reviewed, the following guiding principles are proposed to support responsible, effective, and context-sensitive implementation of AI and Big Data in the public sector: transparency (make algorithmic processes auditable and explainable), equity (identify and mitigate biases and exclusion in data systems), proportionality (align technological solutions with actual policy needs), participation (involve citizens and civil society in the design and evaluation of systems), sustainability (ensure long-term viability, maintenance, and adaptability of digital infrastructures), and sovereignty (protect national and community control over critical data assets).

6.3. Policy recommendations

Governments should develop national AI and big data strategies linked to long-term public value goals; invest in public-sector digital literacy and interdisciplinary training; establish legal frameworks for data protection, algorithmic transparency, and ethical auditing; and promote interagency interoperability through open standards and shared infrastructure.

Academic and research institutions should conduct comparative and longitudinal studies on the impact of AI in governance; expand methodological toolkits to include simulation, ethnography, and participatory research; create translational research platforms to inform real-time policy design; and address knowledge asymmetries by supporting locally rooted scholarship, especially in the Global South.

Private sector and technology providers should design AI tools for interoperability, fairness, and accountability; collaborate with public actors in open innovation environments; respect local norms and data sovereignty frameworks; and contribute to capacity building in the public sector through training and knowledge transfer. Table 7 shows the implementation guidelines for ethical and effective AI and big data use in public policy.

Table 7 Implementation guidelines for ethical and effective AI and big data use in public policy.

Dimension	Guideline	Lead Actors
Technical	Establish interoperable and resilient infrastructure	Government CIOs, IT Departments
Institutional	Integrate training, incentives, and leadership support	Civil service academies, HR units
Ethical	Conduct algorithmic impact assessments	Regulatory bodies, ethics committees
Legal	Enact adaptive data governance legislation	Parliaments, ministries of justice
Civic Engagement	Create participatory mechanisms for oversight	Civil society, citizen councils
Knowledge Sharing	Promote open data and intersectoral learning platforms	Academia, multilateral organizations

Source: Authors’ elaboration on the basis of reviewed literature and institutional best practices (OECD, 2021; Morley et al., 2020; Crawford et al., 2021).

6.4. Final reflection

Big data and AI offer governments the ability to predict outcomes, anticipate needs, optimize interventions, and respond to citizens with greater speed and intelligence. However, their adoption is not determined by technical capability alone; it is equally shaped by behavioral dynamics in human–AI interactions and by citizens’ perceptions of fairness and legitimacy. However, these tools are not neutral. Their deployment reflects choices—technical, political, and ethical—that will shape the contours of governance in the 21st century. A new social contract around digital governance must be built, grounded in transparency, justice, and democratic accountability. Only through such a deliberate and inclusive approach can the potential of these technologies be harnessed for the public good—one that integrates accountability mechanisms, mitigates automation bias, and actively fosters citizen trust.



Ethical Considerations

Not applicable.

Conflict of Interest

The authors declare that they have no conflicts of interest.

Funding

This research did not receive any financial support.

References

- Alon-Barkat, S., & Busuioc, M. (2023). Human–AI interactions in public sector decision making: “Automation bias” and “selective adherence” to algorithmic advice. *Journal of Public Administration Research and Theory*, 33(1), 153–169. <https://doi.org/10.1093/jopart/muac007>
- Alvarez-Aros, E. L., & Bernal-Torres, C. A. (2021). Technological competitiveness and emerging technologies in Industry 4.0 and Industry 5.0. *Anais da Academia Brasileira de Ciências*, 93(1), e20191290. <https://doi.org/10.1590/0001-3765202120191290>
- Amaral, S., Santos, J. R., & Martins, M. (2020). Monitoring deforestation in the Amazon using satellite imagery and machine learning. *Journal of Environmental Sciences*, 58(3), 220–231. <https://doi.org/10.1016/j.jes.2020.03.005>
- Ananny, M., & Crawford, K. (2018). Seeing without knowing: Limitations of the transparency ideal and its application to algorithmic accountability. *New Media & Society*, 20(3), 973–989. <https://doi.org/10.1177/1461444816676645>
- Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016). *Machine bias: There’s software used across the country to predict future criminals. And it’s biased against Blacks*. ProPublica. <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>
- Araujo, T., Helberger, N., Kruijemeier, S., & de Vreese, C. H. (2020). In AI we trust? Perceptions about automated decision-making by artificial intelligence. *AI & Society*, 35(3), 611–623. <https://doi.org/10.1007/s00146-019-00931-w>
- Baller, S., Dutta, S., & Lanvin, B. (2016). *The global information technology report 2016: Innovating in the digital economy*. World Economic Forum. https://www3.weforum.org/docs/GITR2016/WEF_GITR_Full_Report.pdf
- Batty, M. (2013). Big data, smart cities and city planning. *Dialogues in Human Geography*, 3(3), 274–279. <https://doi.org/10.1177/2043820613513390>
- Baykurt, B. (2022). Algorithmic accountability in U.S. cities: Transparency, impact, and political economy. *Big Data & Society*, 9(2), 1–12. <https://doi.org/10.1177/20539517221115426>
- Bernal-Torres, C. A., Torres-Guevara, L. E., Alvarez-Aros, E. L., Barrera-Canto, J. L., & Ojeda-López, R. N. (2025). Collaborative work in the implementation of Industry 4.0 technologies and its relationship with business sustainability in an emerging economy. *International Journal of Innovation Studies*, 9(3), 218–230. <https://doi.org/10.1016/j.ijis.2025.05.002>
- Burrell, J. (2016). How the machine ‘thinks’: Understanding opacity in machine learning algorithms. *Big Data & Society*, 3(1), 1–12. <https://doi.org/10.1177/2053951715622512>
- Choi, H., & Varian, H. (2012). Predicting the present with Google Trends. *Economic Record*, 88(s1), 2–9. <https://doi.org/10.1111/j.1475-4932.2012.00809.x>
- Couldry, N., & Mejias, U. A. (2019). Data colonialism: Rethinking big data’s relation to the contemporary subject. *Television & New Media*, 20(4), 336–349. <https://doi.org/10.1177/1527476418796632>
- Crawford, K., Dobson, K., & MacDonald, F. (2021). Data colonialism and the limits of AI ethics. *Big Data & Society*, 8(2), 1–11. <https://doi.org/10.1177/20539517211017227>
- Das, R., & Zhang, Z. (2019). AI and big data in healthcare: The next frontier for innovation. *Journal of Healthcare Informatics*, 12(3), 45–61.
- Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3), 319–340. <https://doi.org/10.2307/249008>
- Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 4171–4186). Association for Computational Linguistics. <https://doi.org/10.18653/v1/N19-1423>
- Eubanks, V. (2018). *Automating inequality: How high-tech tools profile, police, and punish the poor*. St. Martin’s Press.
- Eyert, F., Irgmaier, F., & Ulbricht, L. (2022). Extending the framework of algorithmic regulation. *Regulation & Governance*, 16(4), 1057–1073. <https://doi.org/10.1111/rego.12371>
- Floridi, L., Cowls, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., ... Vayena, E. (2018). AI4People—An ethical framework for a good AI society: Opportunities, risks, principles, and recommendations. *Minds and Machines*, 28(4), 689–707. <https://doi.org/10.1007/s11023-018-9482-5>
- Gartner. (2019). *Hype cycle for emerging technologies 2019*. Gartner Research. <https://www.gartner.com/en/documents/3981005/hype-cycle-for-emerging-technologies-2019>
- Gil-Garcia, J. R., Pardo, T. A., & Nam, T. (2020). What makes a city smart? Identifying core components and proposing an integrative and comprehensive conceptualization. *Information Policy*, 25(1), 3–20. <https://doi.org/10.3233/IP-190124>
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep learning*. MIT Press.
- Hardoon, D., & Shmueli, G. (2013). *Getting started with business analytics: Insightful decision-making*. CRC Press.
- Hughes, T. P. (1987). The evolution of large technological systems. In W. E. Bijker, T. P. Hughes, & T. J. Pinch (Eds.), *The social construction of technological systems* (pp. 51–82). MIT Press.
- Kitchin, R. (2014). *The data revolution: Big data, open data, data infrastructures & their consequences*. SAGE Publications. <https://doi.org/10.4135/9781473909472>



- Laney, D. (2001). *3D data management: Controlling data volume, velocity, and variety* (META Group Research Note). <https://blogs.gartner.com/doug-laney/files/2012/01/ad949-3D-Data-Management-Controlling-Data-Volume-Velocity-and-Variety.pdf>
- Manyika, J., Chui, M., Brown, B., Bughin, J., Dobbs, R., Roxburgh, C., & Byers, A. H. (2011). *Big data: The next frontier for innovation, competition, and productivity*. McKinsey Global Institute. <https://www.mckinsey.com/business-functions/mckinsey-digital/our-insights/big-data-the-next-frontier-for-innovation>
- Margetts, H., & Dunleavy, P. (2013). The second wave of digital-era governance: A quasi-paradigm for government on the Web. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 371(1987), 20120382. <https://doi.org/10.1098/rsta.2012.0382>
- Mayer-Schönberger, V., & Cukier, K. (2013). *Big data: A revolution that will transform how we live, work, and think*. Houghton Mifflin Harcourt.
- Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K., & Galstyan, A. (2021). A survey on bias and fairness in machine learning. *ACM Computing Surveys*, 54(6), Article 115, 1–35. <https://doi.org/10.1145/3457607>
- Mergel, I., Gong, Y., & Bertot, J. C. (2018). Agile government: Systematic literature review and future research. *Government Information Quarterly*, 35(2), 291–298. <https://doi.org/10.1016/j.giq.2018.04.003>
- Mittelstadt, B. D., Allo, P., Taddeo, M., Wachter, S., & Floridi, L. (2016). The ethics of algorithms: Mapping the debate. *Big Data & Society*, 3(2), 1–21. <https://doi.org/10.1177/2053951716679679>
- Moher, D., Liberati, A., Tetzlaff, J., Altman, D. G., & The PRISMA Group. (2009). Preferred reporting items for systematic reviews and meta-analyses: The PRISMA statement. *PLOS Medicine*, 6(7), e1000097. <https://doi.org/10.1371/journal.pmed.1000097>
- Morley, J., Floridi, L., Kinsey, L., & Elhalal, A. (2020). From what to how: An initial review of publicly available AI ethics tools, methods and research to translate principles into practices. *Science and Engineering Ethics*, 26(4), 2141–2168. <https://doi.org/10.1007/s11948-019-00165-5>
- Muralidharan, K., Niehaus, P., & Sukhtankar, S. (2016). Building state capacity: Evidence from biometric smartcards in India. *American Economic Review*, 106(10), 2895–2929. <https://doi.org/10.1257/aer.20141346>
- Neumann, O., Guirguis, K., & Steiner, R. (2022). Exploring artificial intelligence adoption in public organizations: A comparative case study. *Public Management Review*, 26(1), 114–141. <https://doi.org/10.1080/14719037.2022.2048685>
- North, M. J., & Macal, C. M. (2007). *Managing business complexity: Discovering strategic solutions with agent-based modeling and simulation*. Oxford University Press.
- OECD. (2019). *The path to becoming a data-driven public sector*. OECD Publishing. <https://doi.org/10.1787/059814a7-en>
- OECD. (2021). *Recommendation of the Council on Artificial Intelligence*. OECD Legal Instruments. <https://legalinstruments.oecd.org/en/instruments/OECD-LEGAL-0449>
- Petticrew, M., & Roberts, H. (2006). *Systematic reviews in the social sciences: A practical guide*. Blackwell Publishing.
- Rogers, E. M. (2003). *Diffusion of innovations* (5th ed.). Free Press.
- Russell, S., & Norvig, P. (2021). *Artificial intelligence: A modern approach* (4th ed.). Pearson.
- Sun, T. Q., & Medaglia, R. (2019). Mapping the challenges of artificial intelligence in the public sector: Evidence from public healthcare. *Government Information Quarterly*, 36(2), 368–383. <https://doi.org/10.1016/j.giq.2018.09.008>
- Toots, M., McBride, K., Kalvet, T., & Krimmer, R. (2017). Open data as enabler of public service co-creation: Exploring the drivers and barriers. *Information Polity*, 22(3), 231–247. <https://doi.org/10.3233/IP-170418>
- UNESCO. (2021). *AI and education: Guidance for policy-makers*. UNESCO Publishing. <https://unesdoc.unesco.org/ark:/48223/pf0000376707>
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478. <https://doi.org/10.2307/30036540>
- Wenzelburger, G., König, P. D., Felfeli, J., & Achtziger, A. (2024). Algorithms in the public sector: Why context matters. *Public Administration*, 102(1), 40–60. <https://doi.org/10.1111/padm.12901>
- White, T. (2015). *Hadoop: The definitive guide* (4th ed.). O'Reilly Media.
- Wirtz, B. W., Weyerer, J. C., & Geyer, C. (2019). Artificial intelligence and the public sector—Applications and challenges. *International Journal of Public Administration*, 42(7), 596–615. <https://doi.org/10.1080/01900692.2018.1498103>
- Yeung, K. (2018). Algorithmic regulation: A critical interrogation. *Regulation & Governance*, 12(4), 505–523. <https://doi.org/10.1111/rego.12160>
- Zicari, R. V. (2014). *Ethical and social challenges of data science* (Frankfurt Big Data Lab White Paper). <https://www.frankfurt-bigdata-lab.de/wp-content/uploads/2014/12/Ethical-and-Social-Challenges-of-Data-Science.pdf>
- Zuboff, S. (2019). *The age of surveillance capitalism: The fight for a human future at the new frontier of power*. PublicAffairs.
- Zuiderwijk, A., Chen, Y.-C., & Salem, F. (2021). Implications of the use of artificial intelligence in public governance: A systematic literature review and a research agenda. *Government Information Quarterly*, 38(3), 101577. <https://doi.org/10.1016/j.giq.2021.101577>

